

Post-Quantum Cryptography: NIST FIPS Standards

Day 01 | Category: NIST Standards Overview

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NIST FIPS SPECIAL

1. The Quantum Threat: Why PQC Matters

For decades, modern internet security has relied on public-key cryptosystems such as RSA, Diffie-Hellman (DH), and Elliptic Curve Cryptography (ECC). These algorithms protect everything from web traffic (HTTPS/TLS) and virtual private networks (VPNs) to secure messaging and identity authentication.

However, these systems are built on mathematical problems (integer factorization and discrete logarithms) that are vulnerable to quantum computers. Using **Shor's Algorithm**, a sufficiently large and fault-tolerant quantum computer will be able to break these traditional algorithms in polynomial time, completely undermining the confidentiality and integrity of digital communication.

The "Store Now, Decrypt Later" (SNDL) Attack:

Adversaries are currently intercepting and storing encrypted high-value internet traffic. Even though they cannot decrypt it today, they will do so as soon as a cryptographically relevant quantum computer (CRQC) becomes available. Therefore, migrating to Post-Quantum Cryptography is an urgent priority, not a future luxury.

2. The NIST Standardization Milestone

To counter this threat, the National Institute of Standards and Technology (NIST) initiated a global standardization project in 2016. Out of 82 original candidates, rigorous evaluations led to the publication of the first three official Federal Information Processing Standards (FIPS) on August 13, 2024:

- **FIPS 203 (ML-KEM):** A Module-Lattice-Based Key-Encapsulation Mechanism used to establish shared secrets over public channels (replaces ECDH/DH).
- **FIPS 204 (ML-DSA):** A Module-Lattice-Based Digital Signature Standard used to detect tampering and authenticate identities (replaces RSA/ECDSA).
- **FIPS 205 (SLH-DSA):** A Stateless Hash-Based Digital Signature Standard, serving as an alternative signature scheme built on different mathematical assumptions (hash function

preimage resistance).

3. Deep Dive into the FIPS Standards

A. FIPS 203: ML-KEM (Key-Encapsulation Mechanism)

Derived from the **CRYSTALS-Kyber** algorithm, ML-KEM is designed for secure key exchange. Its mathematical foundation rests on the computational hardness of the **Module Learning with Errors (MLWE)** problem, a lattice-based problem believed to be secure against both classical and quantum adversaries.

ML-KEM provides three parameter sets:

Parameter Set	Equivalent Classical Security	NIST Security Level	Key Applicability
ML-KEM-512	AES-128	Level I	High-performance, lower overhead systems
ML-KEM-768	AES-192	Level III	Recommended default for general internet traffic (TLS)
ML-KEM-1024	AES-256	Level V	Ultra-secure, military-grade long-term protection

B. FIPS 204: ML-DSA (Digital Signature Algorithm)

Derived from **CRYSTALS-Dilithium**, ML-DSA is a primary digital signature standard. Like ML-KEM, its security is rooted in Module-LWE. It is highly efficient and offers a balanced trade-off between signature sizes and execution speed.

- **ML-DSA-44:** (NIST Level II) - Suitable for lightweight applications.
- **ML-DSA-65:** (NIST Level III) - Recommended baseline for general security (comparable to RSA-3072 / ECDSA P-256).
- **ML-DSA-87:** (NIST Level V) - Maximum security for highly critical assets.

C. FIPS 205: SLH-DSA (Stateless Hash-Based Signatures)

Derived from **SPHINCS+**, SLH-DSA is a stateless hash-based signature scheme. Unlike lattice-based schemes, its security depends solely on the strength of underlying hash functions (like SHA-2 or SHA-3). It serves as a vital insurance policy in case mathematical breakthroughs compromise lattice-based cryptography.

Trade-off: SLH-DSA generates significantly larger signatures and requires more computational effort than ML-DSA, but it features incredibly small public keys and highly mature mathematical

security assumptions.

4. Cybersecurity Applicability: Implementing PQC on the Internet

Transitioning the internet to PQC involves updating critical protocols:

1. **TLS 1.3 (Web Traffic):** Integrating ML-KEM in hybrid mode (e.g., X25519 + ML-KEM-768) allows browsers and servers to negotiate quantum-resistant keys while maintaining compatibility with legacy systems.
2. **SSH & VPNs:** Encapsulating tunnel keys using ML-KEM protects administrative connections from SNDL attacks.
3. **Firmware & PKI:** Code signing systems should migrate to digital signatures like ML-DSA or SLH-DSA to prevent unauthorized updates by adversaries with quantum capabilities.

Day 1 Expert Challenge / Auto-Avaliação

1. Why is SLH-DSA considered a crucial safety net for PQC, despite having larger signatures and slower speeds than ML-DSA?
2. Which ML-KEM parameter set is recommended by NIST as the standard baseline for general internet communication?
3. What class of mathematical problems forms the foundation of FIPS 203 and FIPS 204?